Introduction to Panel Data Analysis

Econometrics. Lecture 12

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3. Pooled OLS
4. Fixed effects estimation
5. FE-2SLS estimation
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1. Why panel data?

What is panel?

The same **N** units (people, firms, banks, routes, countries, etc.) observed over **T** time periods.

	$t = 1$	$t = 2$	$t = 3$	$t = 4$	time →
$i = 1$	y_{11}	y_{21}	y_{31}	y_{41}	⋯
$i = 2$	y_{12}	y_{22}	y_{32}	y_{42}	⋯
$i = 3$	y_{13}	y_{23}	y_{33}	y_{43}	⋯
$i = 4$	y_{14}	y_{24}	y_{34}	y_{44}	⋯
units ↓	⋯	⋯	⋯	⋯	

N × T observations, stacked

Panel data are denoted by the use of both *i* and *t* subscripts. This is simply because panel data have **both cross-sectional and time series dimensions:**

$$Y_{it} \quad \text{for } t = 1,2,3,\dots,T \quad \text{and } i = 1,2,3,\dots,N$$

Key distinction with pooled cross sections is that **the same cross-sectional units** are followed over a given time period

Panel data: Example

Data on commercial banks

Period	Bank	Policy rate	Lending rate	NPL	Assets
2013m1	108	35.3	37.1	4.3	451 458
2013m2	108	29.7	40.8	2.5	462 751
⋮	⋮	⋮	⋮	⋮	⋮
2017m12	108	9.7	10.6	21.8	757 523
2013m1	110	35.3	43.6	1.6	47 135
2013m2	110	29.7	48.1	1.7	44 544
⋮	⋮	⋮	⋮	⋮	⋮
2017m12	110	9.7	12.9	4.5	109 398
⋮	⋮	⋮	⋮	⋮	⋮

Limitations of panel data

- More difficult to collect. Following the same units over time is expensive, and non-response compounds across waves.
- Gaps and dropouts are the norm. Firms fail, people leave the survey, a bank gets absorbed. Real panels are usually uneven, and the missing pieces often aren't missing at random.
- Measurement errors: often associated with imprecise questions, incompetence of those conducting surveys, and memory errors.
- Too few time periods, or too few units, and the methods we'll use don't behave well.

Advantages of panel data

- **Control unobserved heterogeneity:** ability, managerial quality, route attractiveness – fixed traits we never observe.
  - *Having multiple observations on the same units allows us to control for certain unobserved characteristics of individuals, firms, and so on*.
- The **sample size can be increased** considerably – much better estimators can be obtained.
- Microdata research avoiding the aggregation problem. Panel data are most often collected at the micro level (observations concern individuals, households, firms).

Panel data in EViews (1)

1. Organise data in Excel

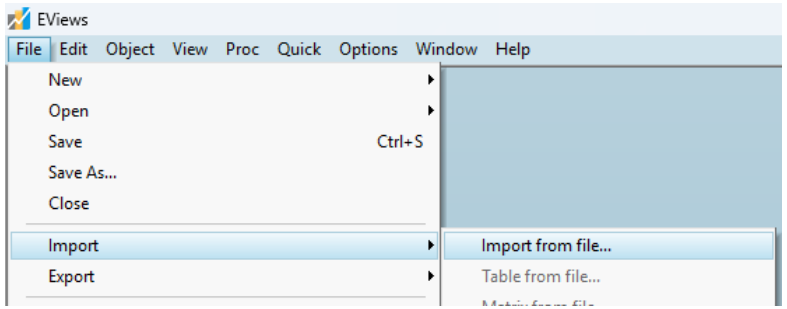
	A	B	C	D	E
1	period	id	price_index	nf	neer
2	01.01.2019	1	1.212637244	1	2.157775557
3	01.02.2019	1	1.210247733	1	2.152253098
4	01.03.2019	1	1.211148892	1	2.125668612
5	01.04.2019	1	1.216178229	1	2.11239183
6	01.05.2019	1	1.216736328	1	2.092252553
7	01.06.2019	1	1.223726109	1	2.066328914
8	01.07.2019	1	1.225538987	1	2.035114172
9	01.08.2019	1	1.22613414	1	2.054237958
10	01.09.2019	1	1.224317546	1	2.074760446
11	01.10.2019	1	1.222661283	1	2.055000098
12	01.11.2019	1	1.225955108	1	2.057914381
13	01.12.2019	1	1.230750877	1	2.106776401
14	01.01.2019	2	1.020384089	1	2.157775557
15	01.02.2019	2	1.024040955	1	2.152253098
16	01.03.2019	2	1.016596193	1	2.125668612
17	01.04.2019	2	1.013455116	1	2.11239183
18	01.05.2019	2	1.012199778	1	2.092252553
19	01.06.2019	2	1.013740872	1	2.066328914
20	01.07.2019	2	1.017072851	1	2.035114172
21	01.08.2019	2	1.021398628	1	2.054237958
22	01.09.2019	2	1.023415636	1	2.074760446
23	01.10.2019	2	1.025749024	1	2.055000098
24	01.11.2019	2	1.03185843	1	2.057914381
25	01.12.2019	2	1.045209829	1	2.106776401
26	01.01.2019	3	1.07502081	0	2.157775557
27	01.02.2019	3	1.070871385	0	2.152253098
28	01.03.2019	3	1.071215323	0	2.125668612
29	01.04.2019	3	1.070817246	0	2.11239183
30	01.05.2019	3	1.076820111	0	2.092252553
31	01.06.2019	3	1.069431953	0	2.066328914
32	01.07.2019	3	1.066377447	0	2.035114172
33	01.08.2019	3	1.068953663	0	2.054237958
34	01.09.2019	3	1.06783211	0	2.074760446
35	01.10.2019	3	1.070981974	0	2.055000098
36	01.11.2019	3	1.077068956	0	2.057914381
37	01.12.2019	3	1.090929298	0	2.106776401

data_panel

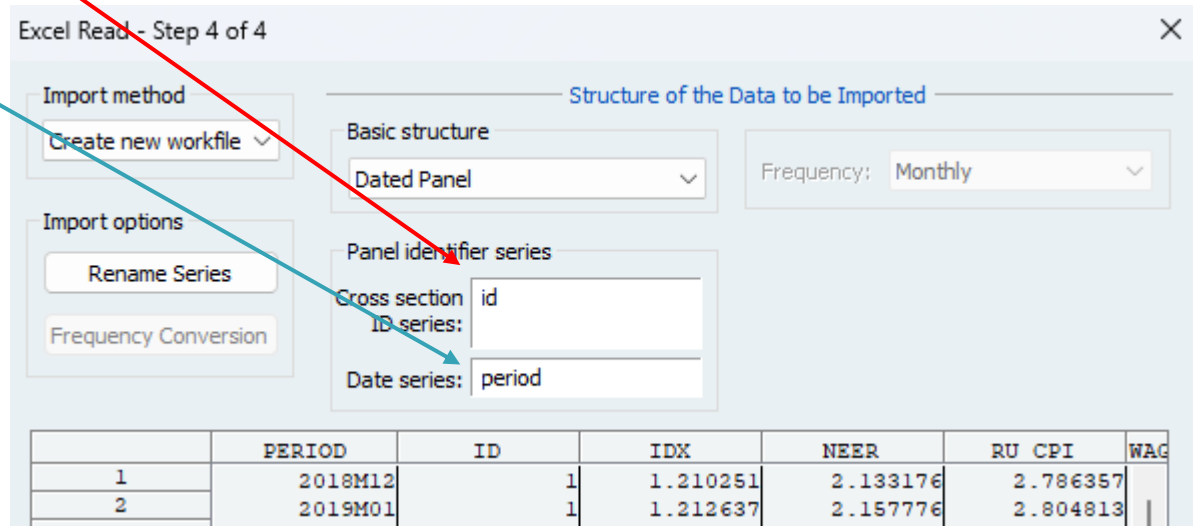
Cross-sections identifier

Time periods identifier

2. Import data to EViews



3. Import data to EViews



Panel data in EViews (2)

Panel data workfile

View | Proc | Object | Save | Snapshot | Freeze | Details+/- | Show | Fetch | Store | Delete | Genr

Range: 2018M12 2026M05 x 49 -- 4410 obs — Filter: *
Sample: 2018M12 2026M05 -- 4410 obs — Order: Name

- c
- id
- idx
- neer
- nf
- period
- prod
- resid
- ru_cpi
- wage

Cross-sections identifier

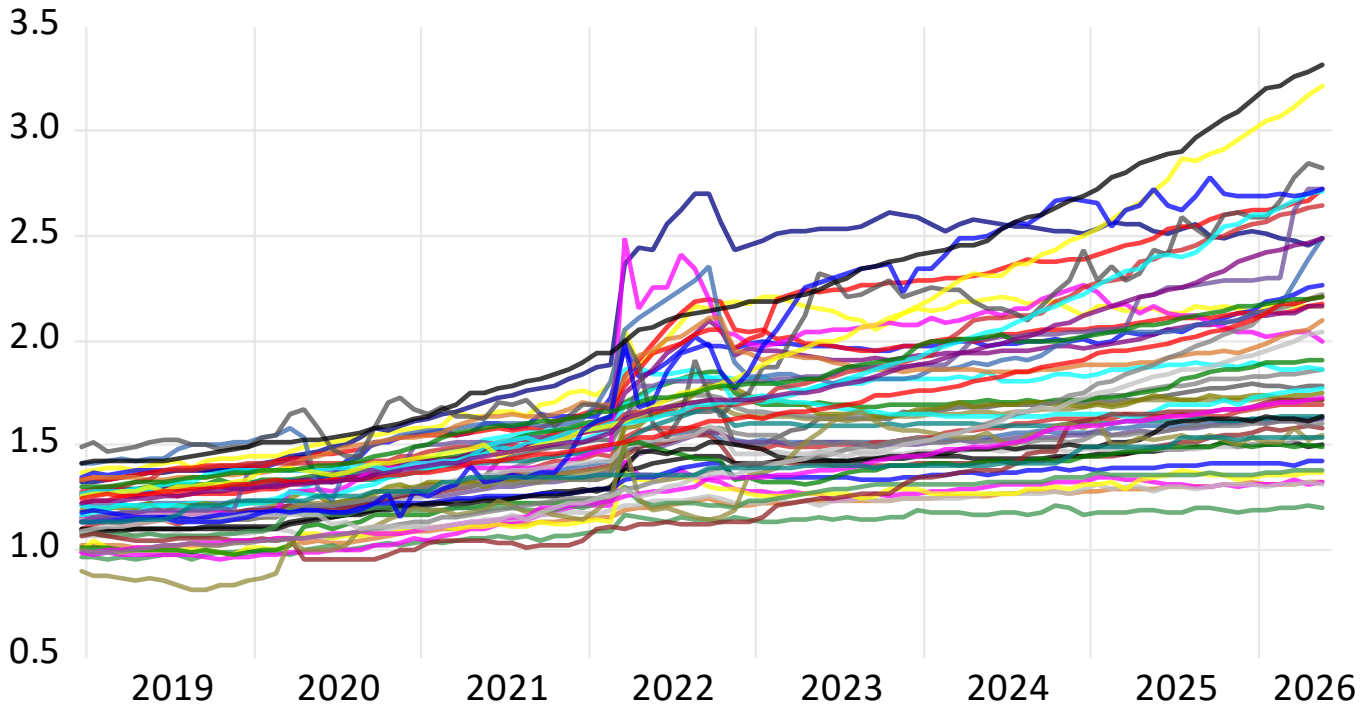
Time periods

Group: UNTITLED Workfile: DATA_ERPT::Data_erpt\

View | Proc | Object | Print | Name | Freeze | Default | Sort | E

	IDX	NEER	NF
1 - 18M12 -	1.210251	2.133176	1
1 - 19M01 -	1.212637	2.157776	1
1 - 19M02 -	1.210248	2.152253	1
1 - 19M03 -	1.211149	2.125669	1
1 - 19M04 -	1.216178	2.112392	1
1 - 19M05 -	1.216736	2.092253	1
1 - 19M06 -	1.223726	2.066329	1
1 - 19M07 -	1.225539	2.035114	1
1 - 19M08 -	1.226134	2.054238	1
1 - 19M09 -	1.224318	2.074760	1
1 - 19M10 -	1.222661	2.055000	1
1 - 19M11 -	1.225955	2.057914	1
1 - 19M12 -	1.230751	2.106776	1

Price indices of individual positions of the consumer basket



2. The unobserved effects model

Unobserved effects

- Significant variables can be omitted from a model primarily for two reasons:
  1. Analyst error
  2. Missing data for the variables.

- It is often much easier to correct analyst error.
- Data may be missing because:
  1. They were not collected for the needs of the given study.
  2. They cannot be collected for methodological reasons

- In both cases, certain phenomena or individual characteristics of the units under study are unobserved and cannot be incorporated into the model.

Linear unobserved effects model

Letting i denote the cross-sectional unit and t the time period, we can write a model for y_{it} with a single observed explanatory variable x_{it} as:

$$y_{it} = \beta_0 + \beta_1 x_{it} + a_i + u_{it} \qquad t = 1, 2, \ldots, T$$

a_i

Unobserved effect

Also: individual heterogeneity, or the fixed effect. Constant over time for unit i, and unobserved.

u_{it}

Idiosyncratic error

Varies over i and t. The usual disturbance; assumed to satisfy the exogeneity conditions.

$v_{it} = a_i + u_{it}$

Composite error

What pooled estimation actually leaves in the error term when a_i is not modelled.

The models we will be looking at will differ in their concept of approaching unobserved effect.

One- and two-way error component models

- **One-way error components model**: the error term v_{it} splits into a unit-specific term a_i and noise u_{it}.

$$v_{it} = a_i + u_{it}$$

- **Two-way error components model**: add a term shared by all units in a period – a common shock or policy change. This is λ_t, indexed by time only – a time effect.

$$v_{it} = a_i + \lambda_t + u_{it}$$

3. Pooled OLS

Pooled OLS

- Ignore a_i entirely: stack all $N \times T$ observations and run OLS on the composite error.

$$y_{it} = \beta_0 + \beta_1 x_{it} + v_{it}, \quad \text{with} \quad v_{it} = a_i + u_{it}$$

Strengths

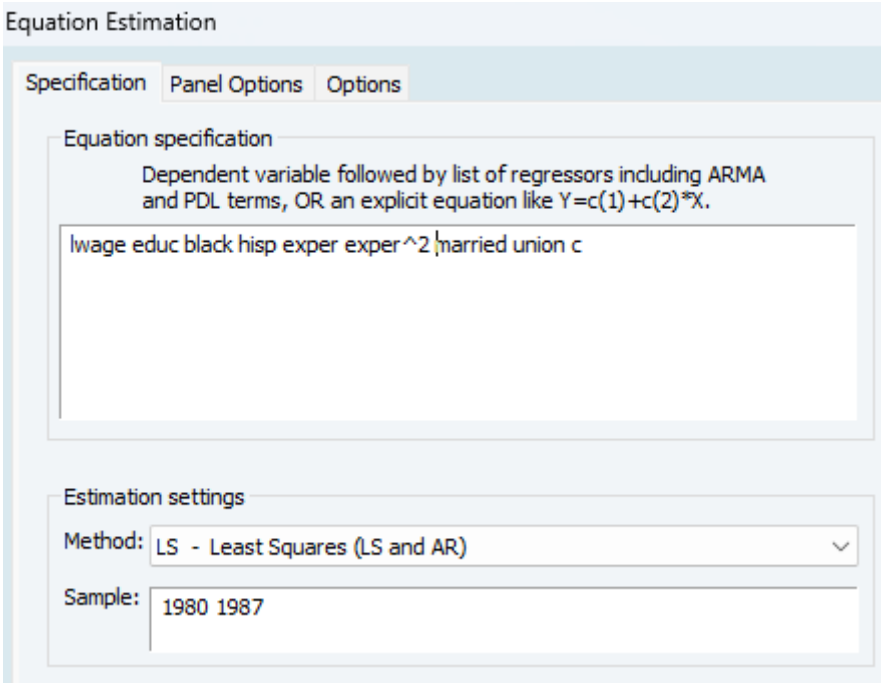
- Simplest possible estimator; a natural baseline.
- Uses both within- and between-unit variation.
- Estimates time-invariant regressors (education, race) without difficulty.

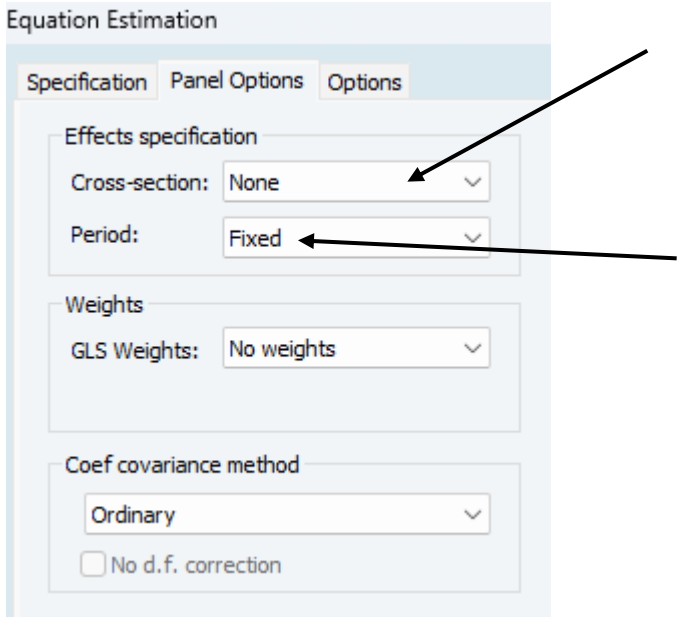
Weaknesses

- Consistent ONLY if Cov(x,a_i)=0 – the strong assumption we wanted to escape.
- a_i sits in v_{it} every period → composite errors are serially correlated.
- Default SEs are therefore wrong.

Pooled OLS in EViews (1)

Data are from Vella and Verbeek (1998). Each of the 545 men in the sample worked in every year from 1980 through 1987. We use the data to estimate a wage equation for men.





We estimate a common constant – no individual effects

We include time effects to account for the common factors affecting wages over time (equivalent to including time period dummies in the regression)

Pooled OLS in EViews (2)

Dependent Variable: LWAGE
Method: Panel Least Squares
Date: 08/05/26 Time: 12:12
Sample: 1980 1987
Periods included: 8
Cross-sections included: 545
Total panel (balanced) observations: 4360

Variable	Coefficient	Std. Error	t-Statistic	Prob.
EDUC	0.091350	0.005237	17.44190	0.0000
BLACK	-0.139234	0.023580	-5.904869	0.0000
HISP	0.016020	0.020797	0.770275	0.4412
EXPER	0.067234	0.013695	4.909478	0.0000
EXPER^2	-0.002412	0.000820	-2.941264	0.0033
MARRIED	0.108253	0.015689	6.899743	0.0000
UNION	0.182461	0.017157	10.63494	0.0000
C	0.179382	0.093227	1.924142	0.0544

Effects Specification

Period fixed (dummy variables)

Root MSE	0.479507	R-squared	0.189278
Mean dependent var	1.649147	Adjusted R-squared	0.186666
S.D. dependent var	0.532609	S.E. of regression	0.480334
Akaike info criterion	1.374764	Sum squared resid	1002.481
Schwarz criterion	1.396714	Log likelihood	-2981.986
Hannan-Quinn criter.	1.382511	F-statistic	72.45876
Durbin-Watson stat	0.864696	Prob(F-statistic)	0.000000

Using Pooled OLS we ignore individual unobserved effects (e.g., motivation, social network, etc.).

To account for individual effects, we should apply different estimators.

4. Fixed effects estimation

Within transformation

1	*Unobserved model*	$y_{it} = \beta_1 x_{it} + a_i + u_{it}$
2	*Average over t for each i*	$\bar{y}_i = \beta_1 \bar{x}_i + a_i + \bar{u}_i$
3	*Subtract (1 – 2):* a_i *cancels*	$(y_{it} - \bar{y}_i) = \beta_1 (x_{it} - \bar{x}_i) + (u_{it} - \bar{u}_i)$

Adding more explanatory variables to the equation causes few changes.

We simply use the time-demeaning on each explanatory variable and then do a pooled OLS regression using all time-demeaned variables.

The unobserved individual effect a_i has disappeared.

Estimate the demeaned equation by pooled OLS – this is the **fixed-effects (within) estimator.**

Fixed effects estimation key assumptions

- **Strict exogeneity**: for the within estimator to be consistent the idiosyncratic error u_{it} should be uncorrelated with each explanatory variable across *all* time periods:

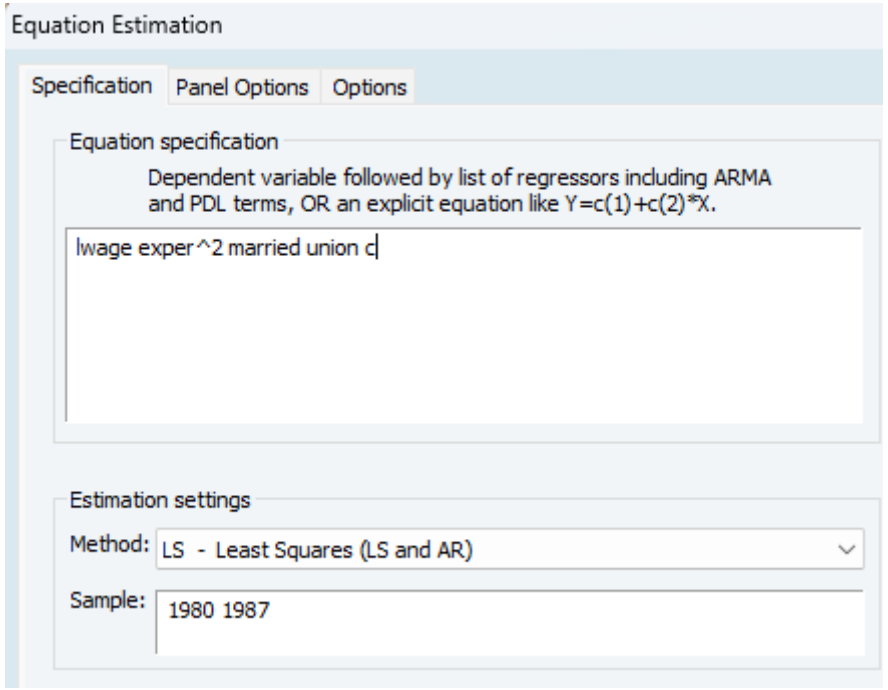
$$\mathrm{E}(u_{it}|x_{i1}, x_{i2}, \ldots, x_{iT}, a_i) = 0 \quad \text{for all } t$$

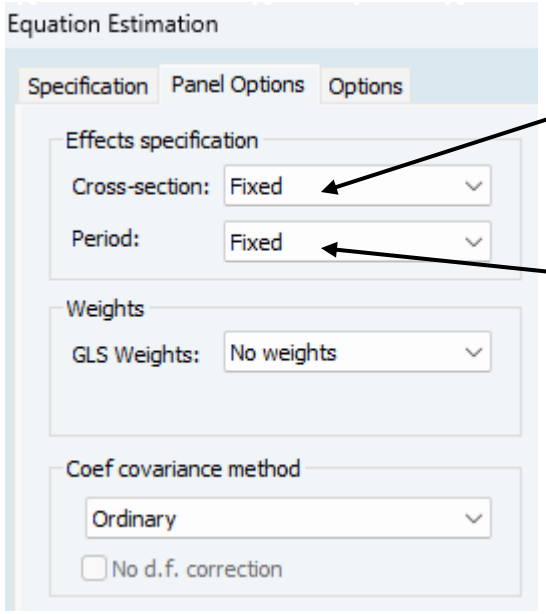
- The **errors** u_{it} are **homoskedastic** and **serially uncorrelated** (across *t*): for the within estimator to be efficient.

- The fixed effects estimator **allows for correlation between a_i and the explanatory variables** in any time period. No assumption on Cov(x, a_i) is needed.

Fixed effects estimation in EViews (1)

Data are from Vella and Verbeek (1998). Each of the 545 men in the sample worked in every year from 1980 through 1987. We use the data to estimate a wage equation for men.





We allow for individual effects (unit-specific constants).

We include time effects to account for the common factors affecting wages over time (equivalent to including time period dummies in the regression).

Fixed effects estimation in EViews (2)

Fixed effects

Dependent Variable: LWAGE
Method: Panel Least Squares
Date: 08/05/26 Time: 12:52
Sample: 1980 1987
Periods included: 8
Cross-sections included: 545
Total panel (balanced) observations: 4360

Variable	Coefficient	Std. Error	t-Statistic	Prob.
EXPER^2	-0.005185	0.000704	-7.361196	0.0000
MARRIED	0.046680	0.018310	2.549385	0.0108
UNION	0.080002	0.019310	4.142962	0.0000
C	1.870609	0.037844	49.42997	0.0000

Effects Specification

Cross-section fixed (dummy variables)
Period fixed (dummy variables)

Root MSE	0.327891	R-squared	0.620912
Mean dependent var	1.649147	Adjusted R-squared	0.565718
S.D. dependent var	0.532609	S.E. of regression	0.350990
Akaike info criterion	0.862313	Sum squared resid	468.7531
Schwarz criterion	1.674475	Log likelihood	-1324.843
Hannan-Quinn criter.	1.148946	F-statistic	11.24956
Durbin-Watson stat	1.821184	Prob(F-statistic)	0.000000

Why racial dummies and education (the same for all periods) are not included in fixed effects estimation?

What about experience?

Marriage and union premiums roughly halve – men who marry or unionise differ in stable, unobserved ways.

EViews recovers the individual effects and normalises them – forcing them to sum to zero – so the overall intercept is actually the average of the individual-specific intercepts

Pooled OLS

Dependent Variable: LWAGE
Method: Panel Least Squares
Date: 08/05/26 Time: 12:12
Sample: 1980 1987
Periods included: 8
Cross-sections included: 545
Total panel (balanced) observations: 4360

Variable	Coefficient	Std. Error	t-Statistic	Prob.
EDUC	0.091350	0.005237	17.44190	0.0000
BLACK	-0.139234	0.023580	-5.904869	0.0000
HISP	0.016020	0.020797	0.770275	0.4412
EXPER	0.067234	0.013695	4.909478	0.0000
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Effects Specification

Period fixed (dummy variables)

Root MSE	0.479507	R-squared	0.189278
Mean dependent var	1.649147	Adjusted R-squared	0.186666
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Durbin-Watson stat	0.864696	Prob(F-statistic)	0.000000

Individual effects explain a large share of wage variation

Time-constant variables and fixed effects (1)

Fixed effects essentially captures all effects that are specific to a particular unit and **do not vary over time**. If we had a panel of countries, the fixed effects would take full account of things such as geographical factors, natural endowments and any other of the many basic factors that vary between countries but not over time.

This means we **cannot directly add variables** that **do not vary over time**, such as country size, etc., as this variable will be perfectly co-linear with the fixed effect.

Experience in our example rises by exactly one each year, so with both individual and time effects present it's perfectly collinear with them and drops too.

Time-constant variables **can be included** in fixed effects estimation as **interactions** with variables that change over time.

Time-constant variables and fixed effects (2)

Has the return to education changed over time?

Dependent Variable: LWAGE
Method: Panel Least Squares
Date: 08/05/26 Time: 13:28
Sample: 1980 1987
Periods included: 8
Cross-sections included: 545
Total panel (balanced) observations: 4360

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	1.396509	0.095729	14.58820	0.0000
MARRIED	0.054820	0.018413	2.977339	0.0029
UNION	0.082978	0.019446	4.267104	0.0000
EDUC*D81	0.011585	0.012262	0.944788	0.3448
EDUC*D82	0.014790	0.012264	1.206054	0.2279
EDUC*D83	0.017118	0.012263	1.395888	0.1628
EDUC*D84	0.016584	0.012266	1.352059	0.1764
EDUC*D85	0.023709	0.012274	1.931635	0.0535
EDUC*D86	0.027412	0.012274	2.233366	0.0256
EDUC*D87	0.030433	0.012272	2.479819	0.0132

Effects Specification

Cross-section fixed (dummy variables)
Period fixed (dummy variables)

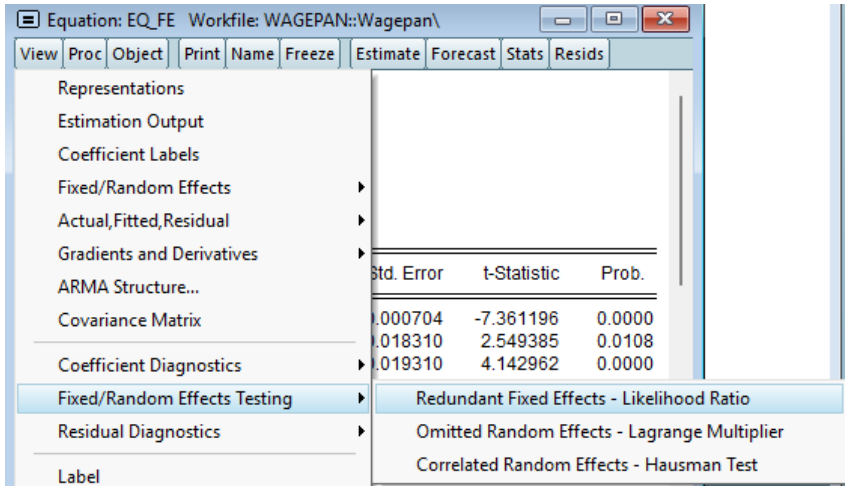
Root MSE	0.329841	R-squared	0.616388
Mean dependent var	1.649147	Adjusted R-squared	0.559841
S.D. dependent var	0.532609	S.E. of regression	0.353357
Akaike info criterion	0.876930	Sum squared resid	474.3479
Schwarz criterion	1.697872	Log likelihood	-1350.708
Hannan-Quinn criter.	1.166662	F-statistic	10.90041
Durbin-Watson stat	1.802453	Prob(F-statistic)	0.000000

The return to education is estimated to be about 3 p.p. larger in 1987 than in the base year, 1980.

However, we do not have an estimate of the return to education in the base year: we cannot use fixed effects to estimate the return to education in the base period, we can only see how the return to education in each year differs from that in the base period.

Pooled OLS vs. fixed effects.
Testing the significance of fixed individual effects and/or time effects

- In the FE model, the presence of fixed effects can be tested.
- The null hypothesis is that **all individual effects** in the model **are zero**.
  - Reject → the effects belong in the model; keep them. Fail to reject → they add nothing, and you can use pooled OLS.



Redundant Fixed Effects Tests
Equation: EQ_FE
Test cross-section and period fixed effects

Effects Test	Statistic	d.f.	Prob.
Cross-section F	9.156772	(544,3805)	0.0000
Cross-section Chi-square	3648.778939	544	0.0000
Period F	29.235537	(7,3805)	0.0000
Period Chi-square	228.410173	7	0.0000
Cross-Section/Period F	10.066772	(551,3805)	0.0000
Cross-Section/Period Chi-square	3920.738468	551	0.0000

We reject the null for both individual and time effects at any level: **both individual and period fixed effects belong in the model**.

First differencing and dummy-variable regression

The unobserved effects model: $y_{it} = \beta_0 + \beta_1 x_{it} + a_i + u_{it}$

1. Within transformation (time-demeaning)

$\bar{y}_i = \beta_0 + \beta_1 \bar{x}_i + a_i + \bar{u}_i \Rightarrow$

$\Rightarrow (y_{it} - \bar{y}_i) = \beta_1 (x_{it} - \bar{x}_i) + (u_{it} - \bar{u}_i)$

Subtract each unit's **own average over all periods**.

Identification comes from how each observation deviates from that unit's mean.

When the u_{it} are serially uncorrelated, fixed effects is more efficient than first differencing.

2. First differencing

$$\Delta y_{it} = \beta_1 \Delta x_{it} + \Delta u_{it},$$

$$\Delta y_{it} = y_{it} - y_{i,t-1}$$

Subtract the **previous period**. As $a_i - a_i = 0$, it drops.

Identification comes from period-to-period changes.

If u_{it} follows a random walk – then the difference Δu_{it} is serially uncorrelated, and first differencing is better.

3. Dummy-variable regression (LSDV)

$$y_{it} = \beta_1 x_{it} + \sum_{j=1}^{N} a_j D_{ji} + u_{it}, D_{ji} = 1 \text{ if } i = j$$

Don't remove a_i – **estimate** it, one intercept per unit via a dummy.

Gives identical $\hat{\beta}_1$ and SEs to time-demeaning.

Infeasible for large N (too many dummies).

First differencing vs. time-demeaning. The effect of worker training on productivity

1. Within transformation (time-demeaning – FE)

Dependent Variable: LSCRAP
Method: Panel Least Squares
Date: 08/05/26 Time: 14:57
Sample: 1987 1989
Periods included: 3
Cross-sections included: 54
Total panel (balanced) observations: 162

Variable	Coefficient	Std. Error	t-Statistic	Prob.
GRANT	-0.252315	0.150629	-1.675075	0.0969
GRANT_1	-0.421589	0.210200	-2.005659	0.0475
C	0.488295	0.060106	8.123876	0.0000

Effects Specification

Cross-section fixed (dummy variables)
Period fixed (dummy variables)

Root MSE	0.398809	R-squared	0.927572
Mean dependent var	0.393681	Adjusted R-squared	0.887876
S.D. dependent var	1.486471	S.E. of regression	0.497744
Akaike info criterion	1.715383	Sum squared resid	25.76593
Schwarz criterion	2.820819	Log likelihood	-80.94602
Hannan-Quinn criter.	2.164207	F-statistic	23.36680
Durbin-Watson stat	1.996983	Prob(F-statistic)	0.000000

FE uses all 3 periods;
FD uses only $T - 1 = 2$

Coefficients are close, but the loss of 1987 inflates SE in FD.

Don't compare R^2s:
- FE – individual effects explain a large share of the scrap rate variation;
- FD – grant changes explain little of scrap changes variation.

2. First differencing (FD)

Dependent Variable: D(LSCRAP)
Method: Panel Least Squares
Date: 08/05/26 Time: 15:01
Sample (adjusted): 1988 1989
Periods included: 2
Cross-sections included: 54
Total panel (balanced) observations: 108

Variable	Coefficient	Std. Error	t-Statistic	Prob.
D(GRANT)	-0.222781	0.130742	-1.703970	0.0914
D(GRANT_1)	-0.351246	0.235085	-1.494124	0.1382
C	-0.138711	0.075184	-1.844958	0.0679

Effects Specification

Period fixed (dummy variables)

Root MSE	0.565935	R-squared	0.036518
Mean dependent var	-0.221132	Adjusted R-squared	0.008725
S.D. dependent var	0.579248	S.E. of regression	0.576716
Akaike info criterion	1.773399	Sum squared resid	34.59049
Schwarz criterion	1.872737	Log likelihood	-91.76352
Hannan-Quinn criter.	1.813677	F-statistic	1.313929
Durbin-Watson stat	1.498132	Prob(F-statistic)	0.273884

Using Holzer et al. (1993) data we want to estimate the effect of the job training grants on worker productivity. *Scrap* is the firm's scrap rate The binary variable *grant* indicates whether the firm received a grant in 1988 for job training.

5. FE-2SLS estimation

Fixed effects removes correlation with the time-invariant effect a_i – but a time-varying regressor can still be correlated with the idiosyncratic error.

Suppose we want to estimate the effect of another hour of job training on worker productivity. For the three years 1987–1989, consider the simple FE model:

$$\log(scrap_{it}) = \beta_0 + \beta_1 hrsemp_{it} + a_i + \lambda_t + u_{it}, \qquad t = 1,2,3.$$

where $scrap_{it}$ is firm i's scrap rate in year t, and $hrsemp_{it}$ is hours of job training per employee. We include both individual (a_i) and time effects (λ_t).

We might worry $hrsemp_{it}$ is correlated with a_i – unmeasured, fixed firm characteristics (management, capital). **Fixed effects removes that.**

But what if u_{it} is correlated with $hrsemp_{it}$ even after the within transformation?

In this case, we need an instrumental variable for demeaned $hrsemp_{it}$.

FE-2SLS.
First stage: Isolating exogenous variation

Receiving a training grant that year ($grant_{it}$) and previous year ($grant_1_{it}$) as instruments for training ($hrsemp_{it}$).

$$\widehat{hrsemp}_{it} = \pi_0 + \pi_1 grant_{it} + \pi_2 grant_1_{it} + c_i + \tau_t$$

Relevance: test $\mathrm{H}_0: \pi_1 = 0, \pi_2 = 0$. We want *F*-stat > 10 to reject the weak instruments.

Exogeneity : Michigan awarded grants roughly first-come, first-served, not on firm performance – grant is plausibly uncorrelated with the firm's scrap shocks u_{it}.

FE-2SLS.
First stage in EViews

The grant effect of +32.8 hours has the right sign and is economically large.

The lagged effect is irrelevant.

Dependent Variable: HRSEMP
Method: Panel Least Squares
Date: 08/06/26 Time: 15:07
Sample: 1987 1989 IF LSCRAP<>NA
Periods included: 3
Cross-sections included: 48
Total panel (unbalanced) observations: 140
White period (cross-section cluster) standard errors & covariance (d.f corrected)
WARNING: estimated coefficient covariance matrix is of reduced rank
Standard error and t-statistic probabilities adjusted for clustering

Variable	Coefficient	Std. Error	t-Statistic	Prob.
GRANT	32.76301	10.10327	3.242813	0.0022
GRANT_1	-8.117338	13.24397	-0.612908	0.5429
C	10.07700	3.237779	3.112319	0.0032

Effects Specification

Cross-section fixed (dummy variables)
Period fixed (dummy variables)

Root MSE	14.77068	R-squared	0.690794
Mean dependent var	15.35192	Adjusted R-squared	0.511595
S.D. dependent var	26.65834	S.E. of regression	18.63044
Akaike info criterion	8.966023	Sum squared resid	30544.22
Schwarz criterion	10.05863	Log likelihood	-575.6216
Hannan-Quinn criter.	9.410027	F-statistic	3.854900
Durbin-Watson stat	2.686744	Prob(F-statistic)	0.000000

Wald Test:
Equation: EQ_FE_IV_1

Test Statistic	Value	df	Probability
F-statistic	9.758503	(2, 88)	0.0001
Chi-square	19.51701	2	0.0001

Null Hypothesis: C(1)=0,C(2)=0
Null Hypothesis Summary:

Normalized Restriction (= 0)	Value	Std. Err.
C(1)	32.76301	10.10327
C(2)	-8.117338	13.24397

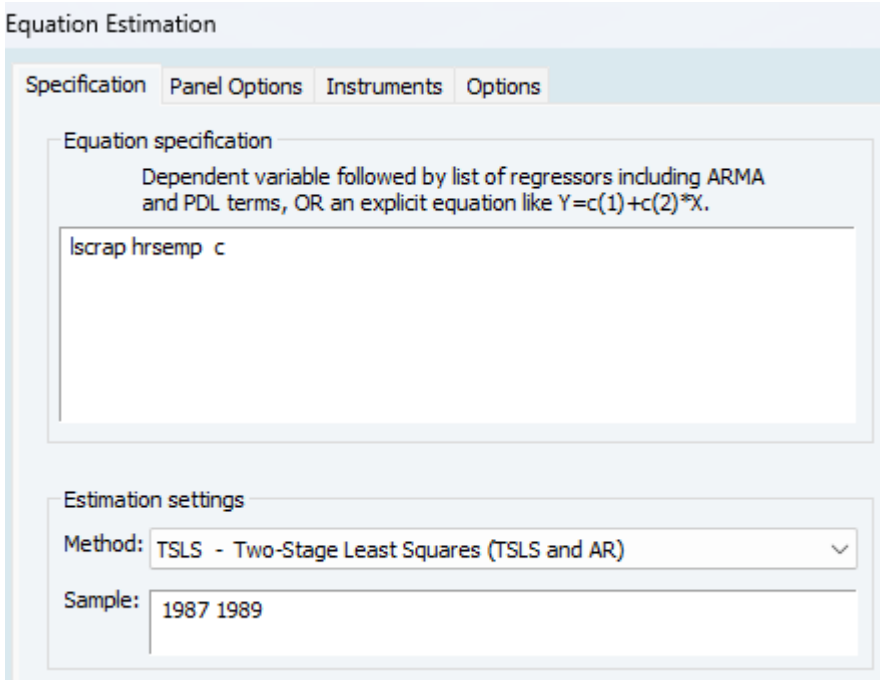
Restrictions are linear in coefficients.

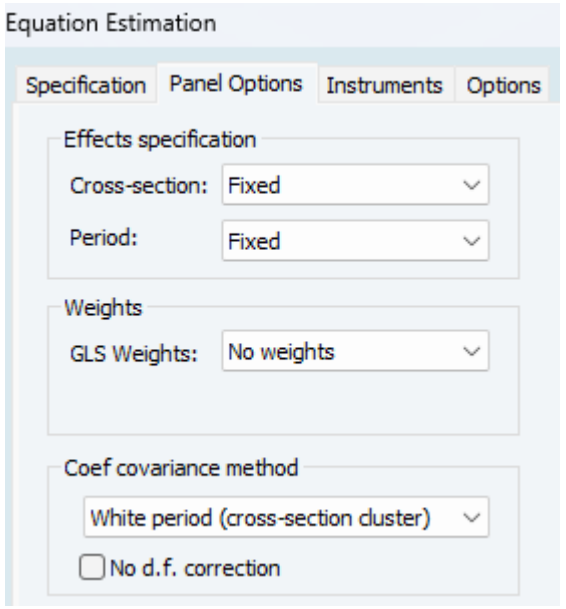
The instruments are not useless as we reject the null. However, *F*-stat is a bit lower than 10. So, instruments are borderline-adequate, not comfortably strong.

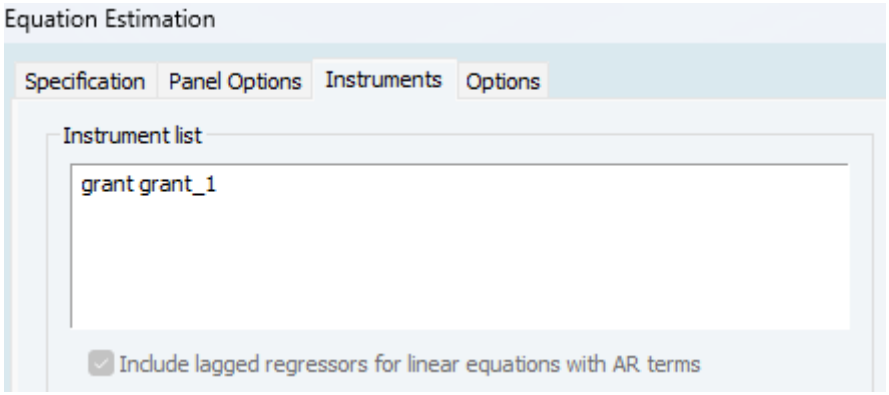
FE-2SLS.
Second stage: Estimation by 2SLS (1)

We use the fitted values ($\widehat{hrsemp}_{it}$) in the equation – only exogenous variation in training:

$$\log(scrap_{it}) = \beta_0 + \beta_1 \widehat{hrsemp}_{it} + a_i + \lambda_t + u_{it}$$







FE-2SLS.
Second stage: Estimation by 2SLS (2)

FE-2SLS

Dependent Variable: LSCRAP
Method: Panel Two-Stage Least Squares
Date: 08/06/26 Time: 15:17
Sample: 1987 1989
Periods included: 3
Cross-sections included: 48
Total panel (unbalanced) observations: 140
White period (cross-section cluster) standard errors & covariance (d.f. corrected)
WARNING: estimated coefficient covariance matrix is of reduced rank
Standard error and t-statistic probabilities adjusted for clustering
Instrument specification: C GRANT GRANT_1
Constant added to instrument list

Variable	Coefficient	Std. Error	t-Statistic	Prob.
HRSEMP	-0.001379	0.002912	-0.473583	0.6380
C	0.436162	0.044704	9.756640	0.0000

Effects Specification

Cross-section fixed (dummy variables)
Period fixed (dummy variables)

Root MSE	0.425149	R-squared	0.914210
Mean dependent var	0.414991	Adjusted R-squared	0.866014
S.D. dependent var	1.456734	S.E. of regression	0.533225
Sum squared resid	25.30523	F-statistic	18.80714
Durbin-Watson stat	1.991306	Prob(F-statistic)	0.000000
Second-Stage SSR	25.50348	Instrument rank	52
Prob(J-statistic)	0.081528		

Two-way FE plus clustering by 48 firms produces more parameters than the clustered covariance can support at full rank. It doesn't bias the coefficients, but it means the SEs and the *J*-stat are not to be trusted to many decimal places.

With 48 firms over three years, our design cannot detect an effect of training hours on the scrap rate.

Testing overidentifying restrictions:

- Reject the null of instruments' exogeneity at 10% level, but not at 5%.
- The instruments' validity is questionable.

A non-rejection in an overidentification test means the instruments are *consistent* with each other, not that they are *valid*. Validity still has to be argued from economics.

6. Random effects estimation

Random effects model

The unobserved effects model:

$$y_{it} = \beta_0 + \beta_1 x_{it1} \ldots + \beta_{\mathrm{k}} x_{itk} + a_i + u_{it},$$

becomes a **random effects model** when we assume that the unobserved effect a_i is uncorrelated with each explanatory variable x_{itj}:

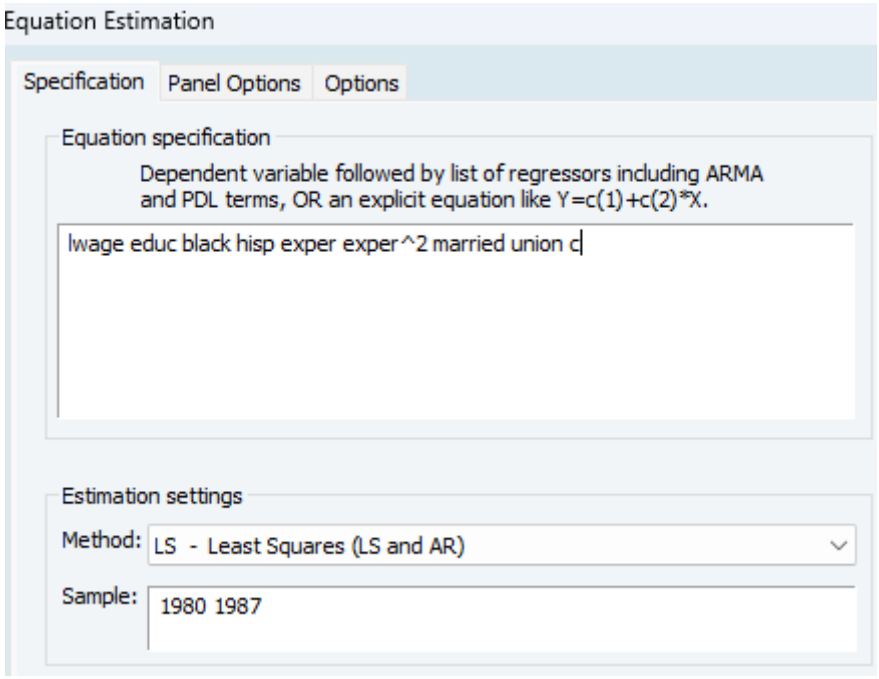
$$\mathrm{Cov}(x_{itj}, a_i) = 0, \qquad t = 1,2, \ldots, T; \;\; j = 1,2, \ldots, k.$$

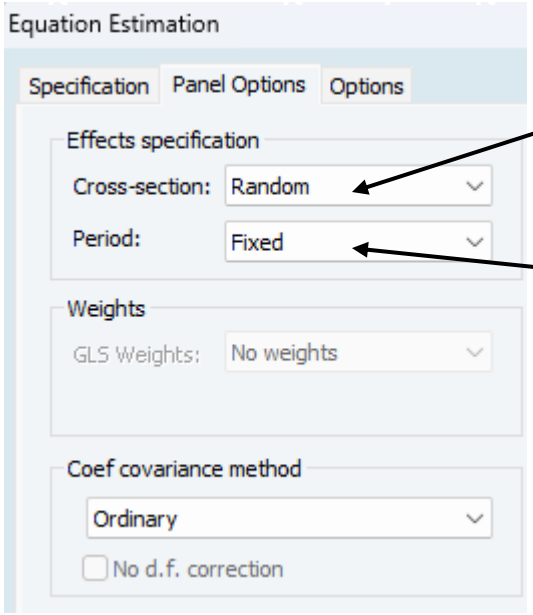
The difference between the fixed effects and the random effects method is that the latter handles the constants for each section not as fixed but as random parameters: a_i is drawn **once per unit** from normal distribution with variance σ_a^2, that *same number* is carried into every period for that unit.

The random effects estimator is the feasible GLS estimator.

Random effects estimation in EViews (1)

Data are from Vella and Verbeek (1998). Each of the 545 men in the sample worked in every year from 1980 through 1987. We use the data to estimate a wage equation for men.





Individual effects are random – not correlated with explanatory variables.

We include time effects to account for the common factors affecting wages over time (equivalent to including time period dummies in the regression).

Pooled OLS vs. Fixed Effects vs. Random Effects

Pooled OLS

Dependent Variable: LWAGE
Method: Panel Least Squares
Date: 08/05/26 Time: 12:12
Sample: 1980 1987
Periods included: 8
Cross-sections included: 545
Total panel (balanced) observations: 4360

Variable	Coefficient	Std. Error	t-Statistic	Prob.
EDUC	0.091350	0.005237	17.44190	0.0000
BLACK	-0.139234	0.023580	-5.904869	0.0000
HISP	0.016020	0.020797	0.770275	0.4412
EXPER	0.067234	0.013695	4.909478	0.0000
EXPER^2	-0.002412	0.000820	-2.941264	0.0033
MARRIED	0.108253	0.015689	6.899743	0.0000
UNION	0.182461	0.017157	10.63494	0.0000
C	0.179382	0.093227	1.924142	0.0544

Effects Specification

Period fixed (dummy variables)

Root MSE	0.479507	R-squared	0.189278
Mean dependent var	1.649147	Adjusted R-squared	0.186666
S.D. dependent var	0.532609	S.E. of regression	0.480334
Akaike info criterion	1.374764	Sum squared resid	1002.481
Schwarz criterion	1.396714	Log likelihood	-2981.986
Hannan-Quinn criter.	1.382511	F-statistic	72.45876
Durbin-Watson stat	0.864696	Prob(F-statistic)	0.000000

Random effects

Dependent Variable: LWAGE
Method: Panel EGLS (Cross-section random effects)
Date: 08/05/26 Time: 16:36
Sample: 1980 1987
Periods included: 8
Cross-sections included: 545
Total panel (balanced) observations: 4360
Swamy and Arora estimator of component variances

Variable	Coefficient	Std. Error	t-Statistic	Prob.
EDUC	0.091876	0.010631	8.642166	0.0000
BLACK	-0.139377	0.047595	-2.928388	0.0034
HISP	0.021732	0.042492	0.511429	0.6091
EXPER	0.105755	0.015326	6.900482	0.0000
EXPER^2	-0.004724	0.000688	-6.869682	0.0000
MARRIED	0.063986	0.016729	3.824781	0.0001
UNION	0.106134	0.017806	5.960582	0.0000
C	0.076021	0.179455	0.423620	0.6719

Effects Specification

	S.D.	Rho
Cross-section random	0.324603	0.4610
Period fixed (dummy variables)		
Idiosyncratic random	0.350990	0.5390

Weighted Statistics

Root MSE	0.351326	R-squared	0.180618
Mean dependent var	1.649147	Adjusted R-squared	0.177977
S.D. dependent var	0.388166	S.E. of regression	0.351932
Sum squared resid	538.1558	F-statistic	68.41243
Durbin-Watson stat	1.589754	Prob(F-statistic)	0.000000

Unweighted Statistics

R-squared	0.182847	Mean dependent var	1.649147
Sum squared resid	1010.433	Durbin-Watson stat	0.846702

Fixed effects

Dependent Variable: LWAGE
Method: Panel Least Squares
Date: 08/05/26 Time: 12:52
Sample: 1980 1987
Periods included: 8
Cross-sections included: 545
Total panel (balanced) observations: 4360

Variable	Coefficient	Std. Error	t-Statistic	Prob.
EXPER^2	-0.005185	0.000704	-7.361196	0.0000
MARRIED	0.046680	0.018310	2.549385	0.0108
UNION	0.080002	0.019310	4.142962	0.0000
C	1.870609	0.037844	49.42997	0.0000

Effects Specification

Cross-section fixed (dummy variables)
Period fixed (dummy variables)

Root MSE	0.327891	R-squared	0.620912
Mean dependent var	1.649147	Adjusted R-squared	0.565718
S.D. dependent var	0.532609	S.E. of regression	0.350990
Akaike info criterion	0.862313	Sum squared resid	468.7531
Schwarz criterion	1.674475	Log likelihood	-1324.843
Hannan-Quinn criter.	1.148946	F-statistic	11.24956
Durbin-Watson stat	1.821184	Prob(F-statistic)	0.000000

Three estimators at a glance

	Pooled OLS	Fixed Effects	Random Effects
Treats a_i as…	part of the error	a parameter (swept out)	a random component
Key assumption	Cov(x, a_i) = 0	none on a_i	Cov(x, a_i) = 0
If Cov(x, a_i) ≠ 0	inconsistent	**consistent**	inconsistent
Variation used	within + between	within only	within + between (weighted)
Time-invariant regressors	estimated	dropped	estimated
Efficiency (if RE holds)	inefficient	consistent, not efficient	efficient (BLUE)
EViews effects setting	Cross-section: None	Cross-section: Fixed	Cross-section: Random

Random vs. fixed effects

Advantages of RE:

- It has fewer parameters to estimate than the fixed effects method.
- It allows for additional explanatory variables that have equal value for all observations within a group (constant over time).

Disadvantages of RE:

- If the unobserved group-specific effects are correlated with the explanatory variables, then the estimators will be biased and inconsistent.

Random vs. fixed effects.
The Hausman test

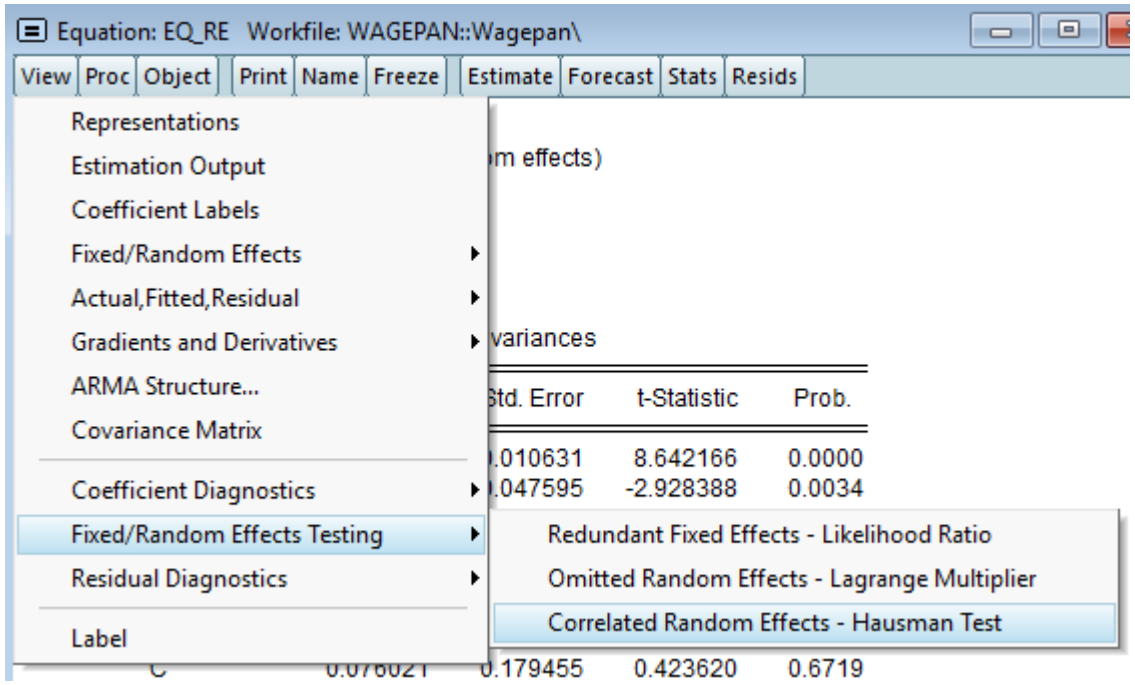
- For the panel data, the appropriate choice between the fixed effects and the random effects methods involves investigating whether the regressors are correlated with the individual (unobserved) effect.
- The Hausman test is formulated to assist in making a choice between the fixed effects and random effects:

H_0: $\text{Cov}(x, a_i) = 0$	H_1: $\text{Cov}(x, a_i) \neq 0$
RE consistent and efficient, FE consistent but inefficient.	RE inconsistent, FE consistent.

- The Hausman statistic may be viewed as a distance measure between the fixed effects and the random effects estimators. Reject H_0 → RE is inconsistent, use fixed effects.

Random vs. fixed effects.
The Hausman test in EViews

Only after FE estimation:



We reject the null: RE is inconsistent

Correlated Random Effects - Hausman Test
Equation: EQ_RE
Test cross-section random effects

Test Summary	Chi-Sq. Statistic	Chi-Sq. d.f.	Prob.
Cross-section random	26.361263	3	0.0000

Cross-section random effects test comparisons:

Variable	Fixed	Random	Var(Diff.)	Prob.
EXPER^2	-0.005185	-0.004724	0.000000	0.0025
MARRIED	0.046680	0.063986	0.000055	0.0201
UNION	0.080002	0.106134	0.000056	0.0005

7. Models’ diagnostics

Four things to check

Is there an unobserved effect at all?

Breusch–Pagan LM test. Reject → heterogeneity matters, do not stop at pooling.

Serial correlation

a_i induces within-unit correlation even under the RE assumption; u_{it} may itself be autocorrelated. Inspect residuals / use a Wooldridge-type test.

Heteroskedasticity

Error variance differs across units (large vs small firms, high-income vs low-income countries). Estimators are still consistent, but inefficient – SE are invalid.

Cross-sectional dependence

Units correlated at a point in time (common shocks). Matters mostly in macro panels with sizeable T.

Getting the standard errors right

The workhorse fix: cluster-robust standard errors, clustered by entity

One correction handles both heteroskedasticity across units and arbitrary serial correlation within a unit. Point estimates are unchanged; only inference is repaired. Use it by default with panel data.

The EViews naming trap

Coef covariance → **White period** ✓

Clusters within a cross-section over time – i.e. entity clustering. This is what you want.

White cross-section ✗

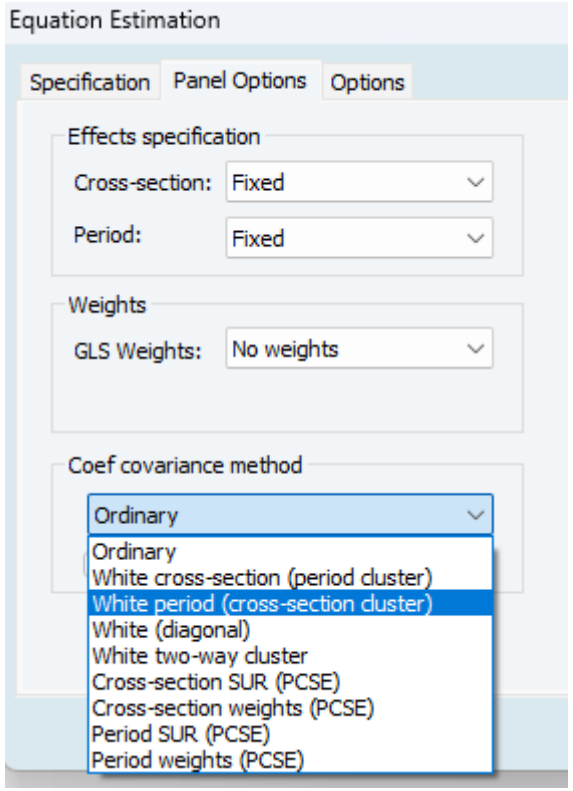
Clusters contemporaneously across units, by period – the opposite.

Two habits worth keeping

Include time dummies by default.
Set period effects to Fixed to absorb aggregate shocks and trends common to all units – cheap insurance Wooldridge recommends routinely.

Match the transformation to the error.
If the idiosyncratic error looks like a random walk, prefer first differencing to FE.

Robust SE in EViews



Fixed effects

Dependent Variable: LWAGE
Method: Panel Least Squares
Date: 08/05/26 Time: 12:52
Sample: 1980 1987
Periods included: 8
Cross-sections included: 545
Total panel (balanced) observations: 4360

Variable	Coefficient	Std. Error	t-Statistic	Prob.
EXPER^2	-0.005185	0.000704	-7.361196	0.0000
MARRIED	0.046680	0.018310	2.549385	0.0108
UNION	0.080002	0.019310	4.142962	0.0000
C	1.870609	0.037844	49.42997	0.0000

Effects Specification

Cross-section fixed (dummy variables)
Period fixed (dummy variables)

Root MSE	0.327891	R-squared	0.620912
Mean dependent var	1.649147	Adjusted R-squared	0.565718
S.D. dependent var	0.532609	S.E. of regression	0.350990
Akaike info criterion	0.862313	Sum squared resid	468.7531
Schwarz criterion	1.674475	Log likelihood	-1324.843
Hannan-Quinn criter.	1.148946	F-statistic	11.24956
Durbin-Watson stat	1.821184	Prob(F-statistic)	0.000000

Fixed effects + robust SE

Dependent Variable: LWAGE
Method: Panel Least Squares
Date: 08/05/26 Time: 20:38
Sample: 1980 1987
Periods included: 8
Cross-sections included: 545
Total panel (balanced) observations: 4360
White period (cross-section cluster) standard errors & covariance (d.f. corrected)
WARNING: estimated coefficient covariance matrix is of reduced rank
Standard error and t-statistic probabilities adjusted for clustering

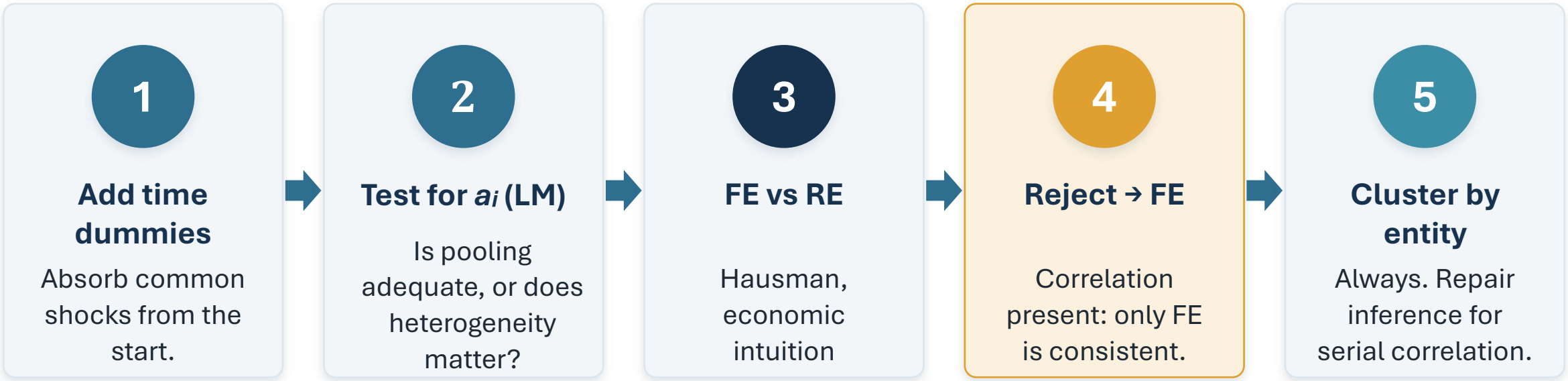
Variable	Coefficient	Std. Error	t-Statistic	Prob.
EXPER^2	-0.005185	0.000925	-5.604835	0.0000
MARRIED	0.046680	0.023984	1.946351	0.0521
UNION	0.080002	0.025970	3.080604	0.0022
C	1.870609	0.049461	37.81985	0.0000

Effects Specification

Cross-section fixed (dummy variables)
Period fixed (dummy variables)

Root MSE	0.327891	R-squared	0.620912
Mean dependent var	1.649147	Adjusted R-squared	0.565718
S.D. dependent var	0.532609	S.E. of regression	0.350990
Akaike info criterion	0.862313	Sum squared resid	468.7531
Schwarz criterion	1.674475	Log likelihood	-1324.843
Hannan-Quinn criter.	1.148946	F-statistic	11.24956
Durbin-Watson stat	1.821184	Prob(F-statistic)	0.000000

A practical decision workflow



Fail to reject at step 3? Use random effects – it is consistent and efficient under that null, and it recovers the coefficients on time-invariant regressors that FE cannot.

Out of scope – and where this goes next

Dynamic panels

A lagged dependent variable makes FE biased (Nickell bias). Handled by GMM estimators.

The bias shrinks as the panel lengthens – negligible once T is large (roughly T ≥ 20–30), non-trivial when T is small (T ≤ 10) and worst at T = 2–3.

Large-T heterogeneous panels

Mean Group and Pooled Mean Group (PMG) estimators, built on ARDL and cointegration. A macro-panel / time-series-panel topic – not needed here.

Panel VARs / ECMs

You can estimate VARs/SVARs on panel data and perform IRFs analysis.

Cross-sectional dependence

Common factors and spatial linkages across units; Pesaran CD test, spatial panels.

8. Difference-in-differences

Difference-in-differences (DiD)

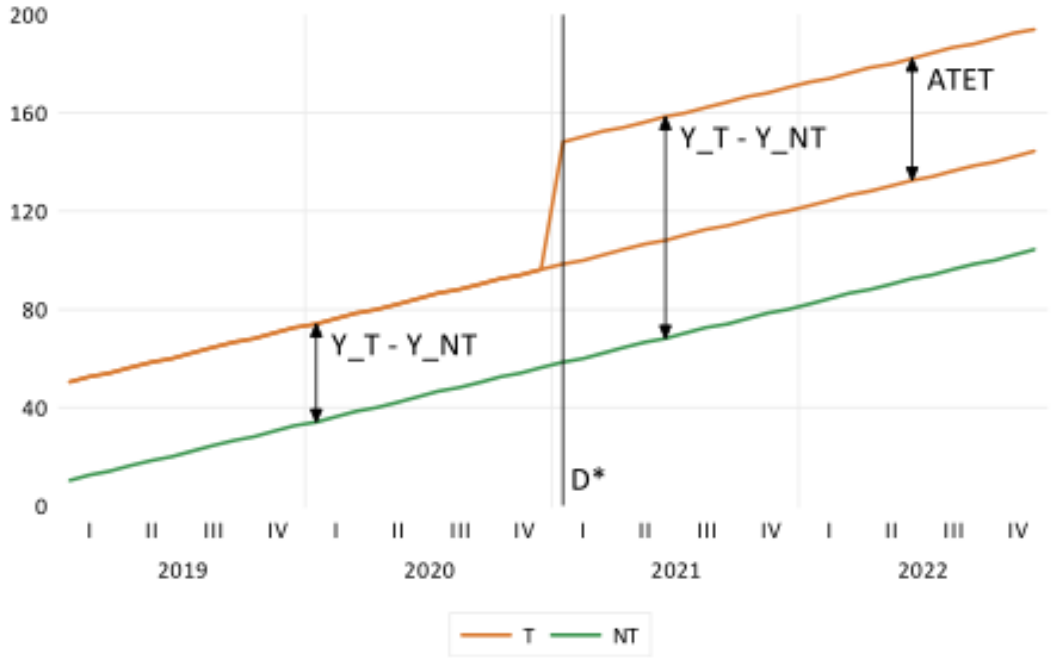
- **Difference-in-difference estimation** is a method of analyzing the impact of a treatment or policy on an outcome by comparing the difference in the outcome variable before and after treatment for those who participated in the treatment, with those who did not participate in the treatment.
  - D^* – a treatment date. *T* – a group of the treated units. *NT* – a group of the untreated units.
  - *Y* – an outcome variable with periods both before and after the treatment date for all units.
- To estimate the average impact of the treatment on *Y*, we compare the difference in the mean of *Y* between the treated and never treated groups before and after the treatment date:

$$\hat{\delta} = (\bar{Y}_{T,t \geq D^*} - \bar{Y}_{NT,t \geq D^*}) - (\bar{Y}_{T,t < D^*} - \bar{Y}_{NT,t < D^*})$$

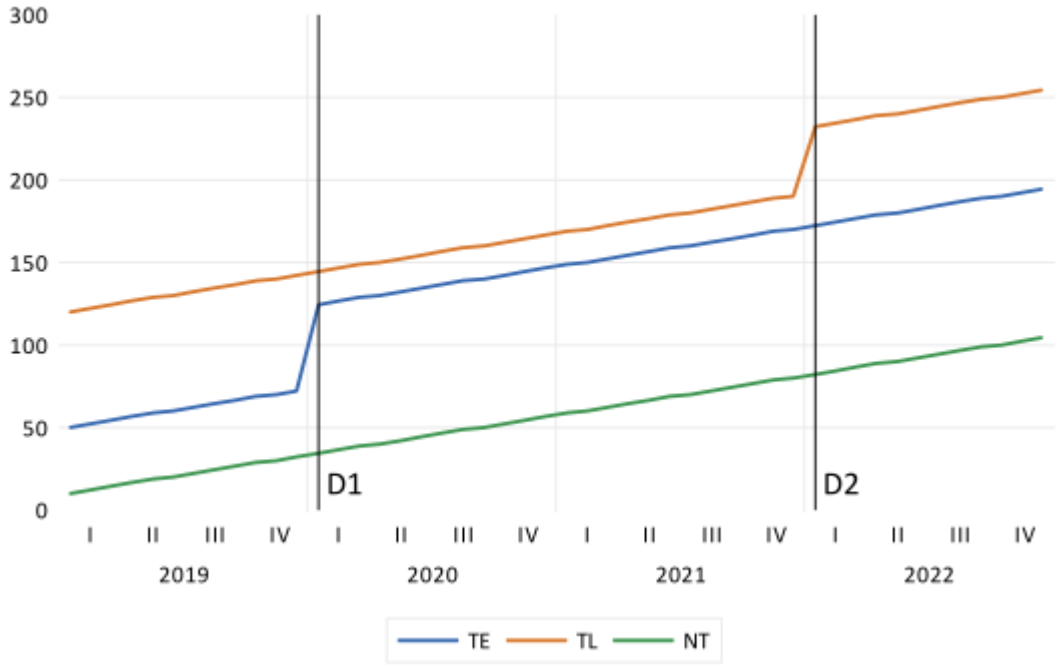
- $\hat{\delta}$ – the Average Treatment Effect for the Treated (ATET).

Difference-in-differences (DiD). Visualization

Single treatment date



Multiple treatment date



The presence of multiple treatment dates does not affect the form of the two-way FE estimator.
We simply extend the two-way FE model to incorporate more than 2 groups.

DiD with panel data

- Consider the panel two-way fixed effects regression:

$$Y_{i,g,t} = \delta D_{g,t} + a_i + \lambda_t + u_{i,g,t}$$

  where g identifies the treatment ($g = 1$) and the non-treatment ($g = 0$) groups, and $D_{g,t}$ takes a value of 1 whenever a treatment group observation is actually treated.

  In this specification, δ – is difference-in-differences estimate.

- **Crucially**, for estimator δ to be unbiased, DiD requires an additional ***parallel trends assumption***: in the counter-factual where individuals in the treated group are not treated, the difference between the treatment group and the control group is constant through time.

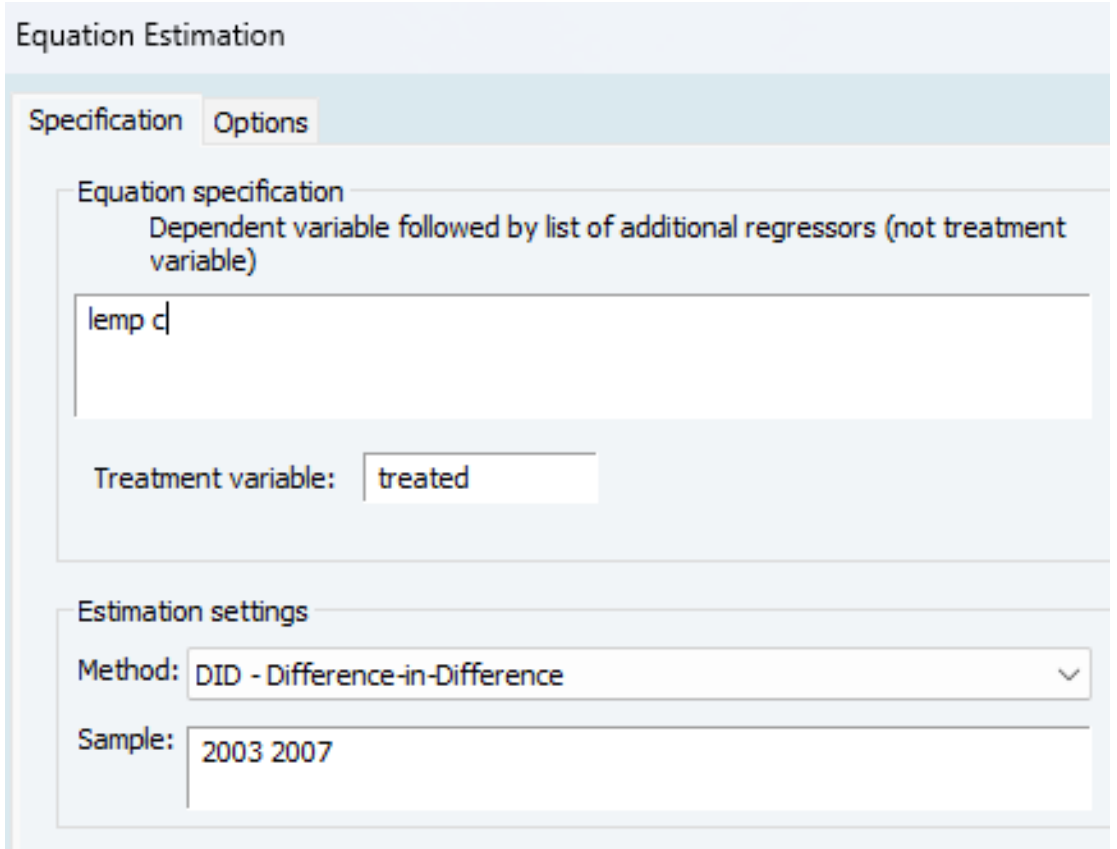
Example.
The impact of the minimum wage on employment

- Callaway and Sant'Anna (2021) study the impact of the minimum wage on employment levels concentrating on teen employment:
  - County level teen employment between 2003–2007.
  - The federal minimum wage remained constant at $5.15 per hour.
  - Some states increased the minimum wage within the state above the federal level during the time period.
  - Counties within such states are the treated group (with different treatment times).
  - Counties within states that did not change their minimum wage are in the control group.
- The series TREATED is a binary variable indicating whether a county has been subject to an increased minimum wage in the time period, the series LEMP contains the log of employment for that county in that time period, and the series LPOP contains the log of the population within the county.

FE-DiD in EViews (1).
The impact of the minimum wage on employment

- To estimate a DiD model in EViews:
  - bring up the equation dialog by clicking on ***Object/New Object.../Equation*** from the main menu bar in your panel workfile.
  - Select ***DiD – Difference-in-Difference*** in the ***Method***.
  - In the ***Equation specification*** field enter the dependent variable followed by any exogenous regressors *apart from the treatment variable*.
  - The *treatment variable* should be entered in the ***Treatment Variable*** edit field.

The treatment series should be a binary variable indicating whether the individual has been treated (i.e., is 1 if the observation in a treatment group which is post-treatment date for that group, and 0 otherwise).



When performing DiD estimation, Eviews uses cross-section cluster-robust standard errors.

FE-DiD in EViews (2).
The impact of the minimum wage on employment

Dependent Variable: LEMP
Method: Difference-in-Difference
Date: 08/06/26 Time: 17:28
Periods included: 5
Cross-sections included: 500
Total panel (balanced) observations: 2500

Variable	Coefficient	Std. Error	t-Statistic	Prob.
TREATED	-0.036549	0.016566	-2.206304	0.0275

R-squared	0.993218	Mean dependent var	5.772516
Adjusted R-squared	0.991505	S.D. dependent var	1.508781
S.E. of regression	0.139060	Akaike info criterion	-0.929474
Sum squared resid	38.57851	Schwarz criterion	0.246984
Log likelihood	1666.842	Hannan-Quinn criter.	-0.502364
F-statistic	579.7363	Durbin-Watson stat	1.686589
Prob(F-statistic)	0.000000	Parallel trend stat	-0.249133
Prob(P. trend)	0.803258		

- Two-way FE-DiD: raising the state minimum wage above the $5.15 federal floor reduced county teen employment by about 3.7% – a statistically significant negative effect.
- Parallel trends is not rejected ($p = 0.80$): treated and control counties were not on significantly different employment paths before the increases.

FE-DiD in EViews (3).
The impact of the minimum wage on employment

DiD

Dependent Variable: LEMP
Method: Difference-in-Difference
Date: 08/06/26 Time: 17:28
Periods included: 5
Cross-sections included: 500
Total panel (balanced) observations: 2500

Variable	Coefficient	Std. Error	t-Statistic	Prob.
TREATED	-0.036549	0.016566	-2.206304	0.0275

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Log likelihood	1666.842	Hannan-Quinn criter.	-0.502364
F-statistic	579.7363	Durbin-Watson stat	1.686589
Prob(F-statistic)	0.000000	Parallel trend stat	-0.249133
Prob(P. trend)	0.803258		

The DiD and manual two-way FE are the *same regression*

Built-in two-way FE estimator

Dependent Variable: LEMP
Method: Panel Least Squares
Date: 08/06/26 Time: 16:36
Sample: 2003 2007
Periods included: 5
Cross-sections included: 500
Total panel (balanced) observations: 2500
White period (cross-section cluster) standard errors & covariance (d.f. corrected)
WARNING: estimated coefficient covariance matrix is of reduced rank
Standard error and t-statistic probabilities adjusted for clustering

Variable	Coefficient	Std. Error	t-Statistic	Prob.
TREATED	-0.036549	0.016566	-2.206304	0.0278
C	5.776771	0.001928	2995.869	0.0000

Effects Specification

Cross-section fixed (dummy variables)
Period fixed (dummy variables)

R-squared	0.993218	Mean dependent var	5.772516
Adjusted R-squared	0.991505	S.D. dependent var	1.508781
S.E. of regression	0.139060	Akaike info criterion	-0.929474
Sum squared resid	38.57851	Schwarz criterion	0.246984
Log likelihood	1666.842	Hannan-Quinn criter.	-0.502364
F-statistic	579.7363	Durbin-Watson stat	1.686589
Prob(F-statistic)	0.000000		

Difference-in-differences.
Other considerations

- You can include additional exogenous regressors. Their inclusion does not change the behavior of the DiD estimator.
- The Group-Time Average Treatment Effects (Callaway-Sant'Anna). This estimator is robust to the treatment effect changing as time from treatment increases.
  - EViews: View/Difference-in-Difference Diagnostics/CS Group-Time Effects…
- Recent research has noted that the TWFE model is not suitable in multiple-timing DiD models if the impact of treatment changes as time from treatment increases. Goodman-Bacon (2021) calculates the treatment effects for individual pairs of treatment groups.
  - EViews: *View/Difference-in-Difference Diagnostics/Goodman-Bacon decomposition*
- Borusyak, Jaravel, and Spiess Imputed Estimator.
  - Eviews:View/Difference-in-Difference Diagnostics/BJS Imputed Estimator

Appendix. The correlated random effects (CRE) approach

One regression that does it all

- Run RE, but add the time-averages of the time-varying regressors as extra controls.

$$y_{it} = \alpha + \beta x_{it} + \gamma \bar{x}_i + r_i + u_{it}, \quad a_i = \alpha + \gamma \bar{x}_i + r_i$$

$\bar{x}_i$ = *the within-unit time average*

The Hausman test is a Wald test

A joint test of H_0: $\gamma = 0$ on the time-averages is the Hausman test – same null, but an ordinary (cluster-robust) Wald test.

$\hat{\beta}$ equals the FE estimate

The slope on x_{it} in this CRE regression is numerically identical to fixed effects. One equation shows the FE–RE link.

γ names the culprits

A significant γ on a given $\bar{x}_i$ tells you which regressor is correlated with a_i .

Allow for time-constant explanatory variables

CRE provides a way to include time-constant explanatory variables in what is effectively a fixed effects analysis.

Doing CRE in EViews

1 Build the time-averages

In a panel workfile, average each regressor within the cross-section id.

2 Add them to the RE equation

Estimate RE with the original regressors plus their within-means.

3 Wald-test the means, robustly

Jointly test the mean coefficients = 0 with a cluster-robust covariance.

Check: the coefficient on concen equals the fixed-effects estimate exactly.

AIRFARE (Wooldridge): a balanced panel of 1,149 U.S. air routes over 1997–2000, with average fare (*fare*), market concentration (*concen*), distance (*dist*), and endpoint populations.

EViews commands

```
' time-average of concen (by route)
series concenbar = @meansby(concen, @crossid)

' random-effects equation with the mean
equation eq_cre.ls(cx=r, cov=perwhite) log(fare) c
concen concenbar

' robust Hausman: joint Wald on the mean(s)
eq_cre.wald  c(3)=0
```

Set the coefficient covariance to White period (entity-clustered) so the Wald test is robust (cov=perwhite).